rank_weighted

equity strategy | family: equity-baseline | status: validated

What this strategy does

All 10 deciles, continuous SUE-rank weight

Why it should work. Same drift as Strategy 1 but harvested across the whole cross-section for lower variance.

Design notes (from the strategy's source file).

Strategy 2 -- Rank-weighted.

Trades every decile with weight proportional to SUE-rank distance from the median: broader diversification than the extreme-decile book, at the price of ~4x more positions (which saturates the liquidity and leverage caps and dilutes the high-conviction names).

parameter	value
base_unit_fraction	0.002
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
beta_overlay	0.0

Performance

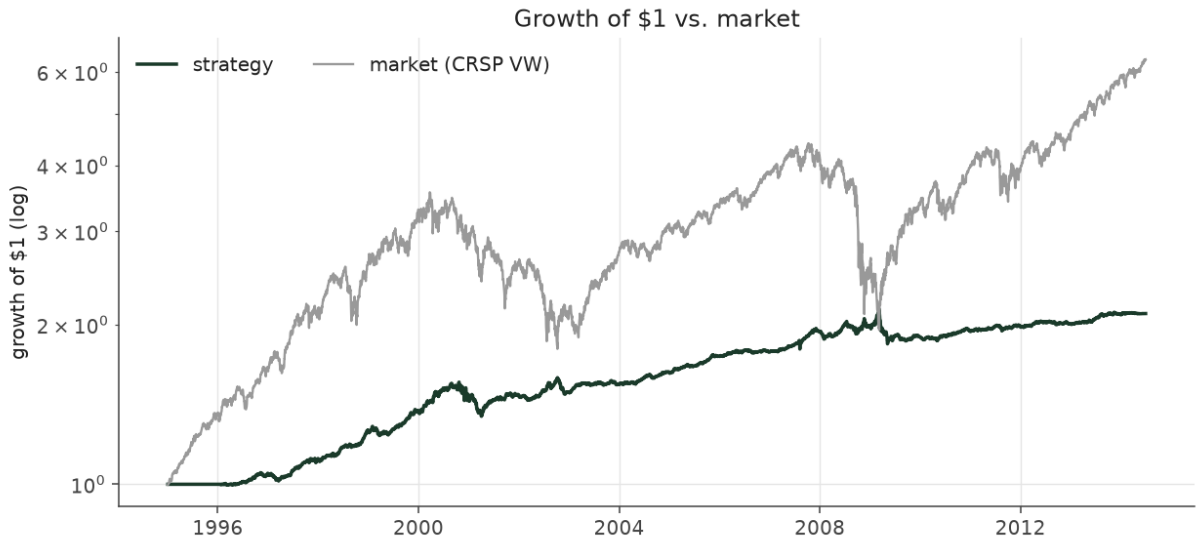
metric	rank weighted	market (CRSP VW)
Annualized return	3.88%	9.94%
Annualized volatility	3.86%	19.35%
Sharpe (NAV-pct)	1.01	0.59
Sortino	1.42	
Max drawdown	-14.0%	-55.5%
Calmar	0.28	
Final NAV ($10M start)	$20,996,808	
Worst day	-2.34%	
% positive days	52.4%	
Sharpe 1996-2001	1.39	
Sharpe 2002-2007	1.55	
Sharpe 2008-2013	0.32	

Correlation of daily returns with the market: **0.01**.

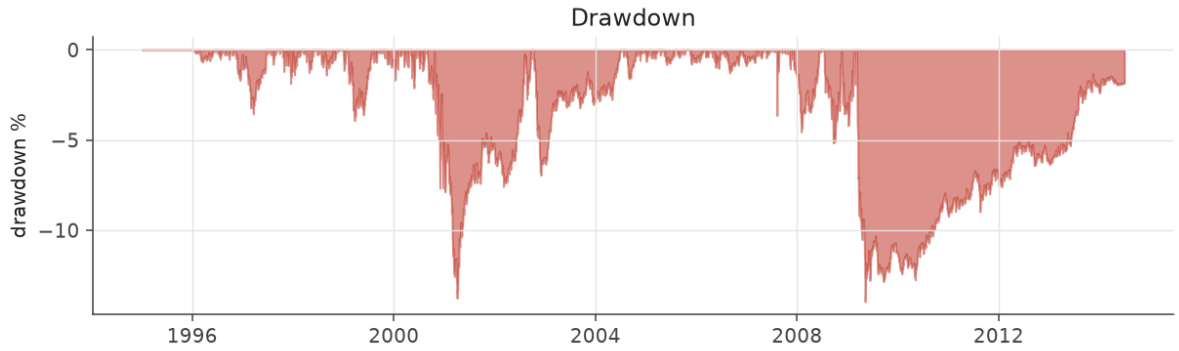
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.01

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0388

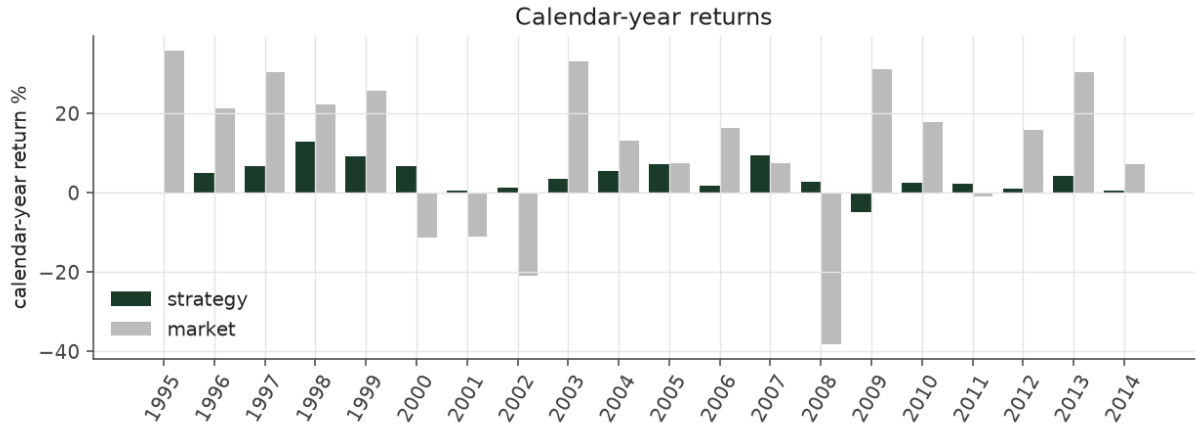
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.278



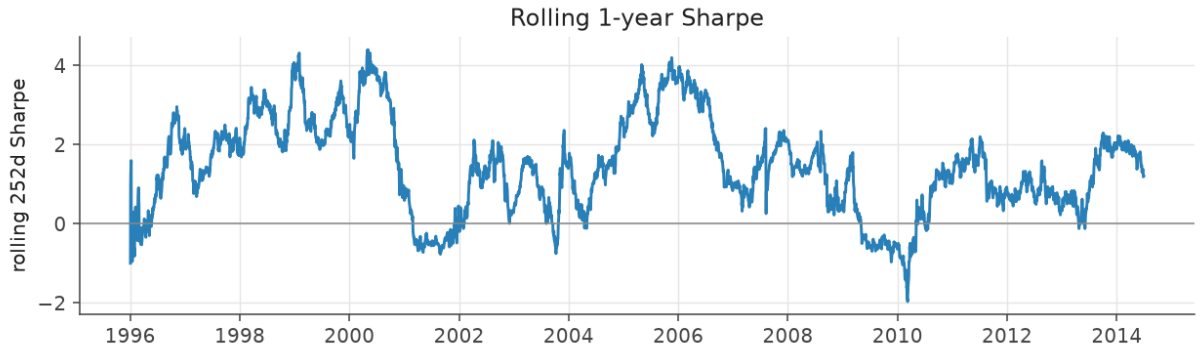
NAV (log scale)



Drawdown from running peak

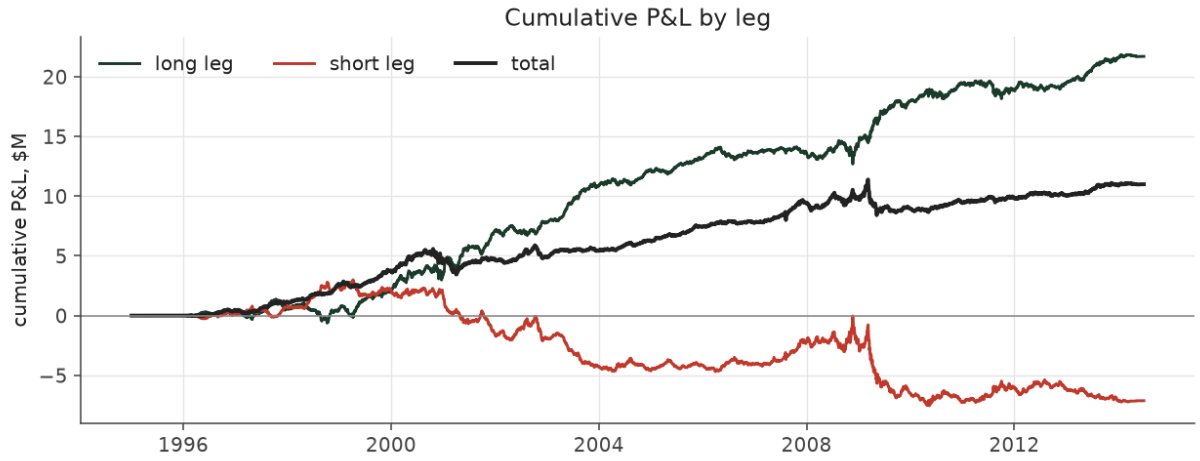


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	21.70	-7.13
P&L after costs ($M)	20.00	-9.00
of which common (average-event) component ($M)	14.40	-13.84
of which excess / PEAD component ($M)	7.30	6.72
excess minus costs ($M)	5.61	4.84
gross return per $ traded (bp)	248.49	-83.56
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	1.19	-0.33

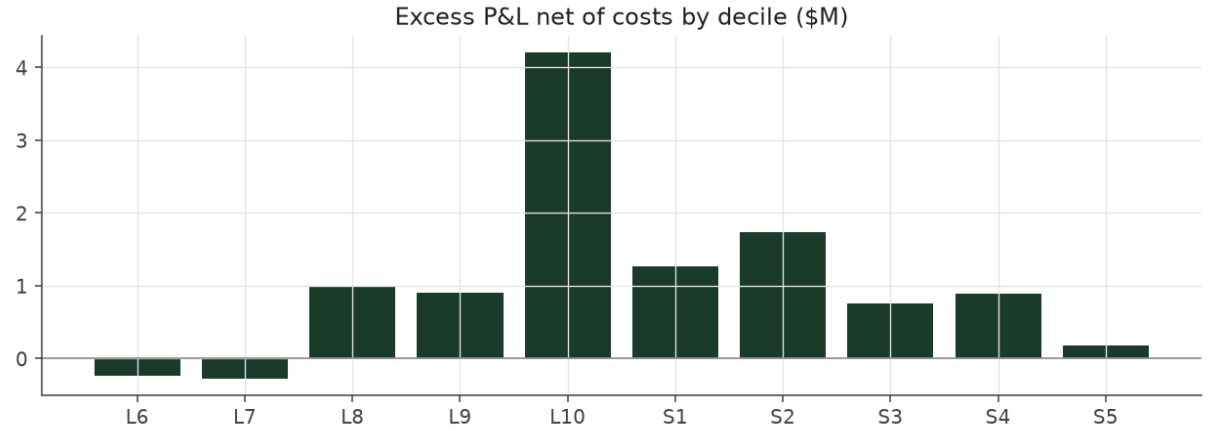
Daily P&L correlation between the two legs: **-0.79**.

• **The short leg loses money in absolute terms** (-7.1M before costs) while the long leg earns +21.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 14.4M / 13.8M. Stripping that common component, the surprise signal itself earned +7.3M on the long side and +6.7M on the short side.

• The two legs' daily P&L are -0.79 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.19 long / -0.33 short vs 1.01 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	6	18,256	41.69	0.34	-0.14	0.09	-0.23	-2.2%	-55.42
long	7	17,431	104.34	1.62	-0.07	0.20	-0.28	-2.7%	-26.66
long	8	16,588	169.97	3.99	1.34	0.33	1.00	9.6%	59.08
long	9	17,747	248.03	4.65	1.38	0.47	0.90	8.7%	36.43
long	10	17,174	309.23	9.39	4.81	0.60	4.21	40.3%	136.06
short	1	17,169	295.94	-3.37	1.94	0.68	1.26	12.1%	42.72
short	2	17,281	238.00	-2.19	2.27	0.52	1.74	16.7%	73.23
short	3	15,970	163.81	-1.95	1.12	0.36	0.76	7.3%	46.33
short	4	21,469	125.93	-1.12	1.16	0.26	0.90	8.6%	71.17
short	5	14,319	29.09	-0.37	0.24	0.06	0.18	1.7%	60.57



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,578	167.94	1.68	3.12	0.68	2.44	23.4%	145.50
2	32,212	338.94	3.21	4.54	0.98	3.56	34.0%	104.91
3	36,915	385.67	3.11	3.66	0.81	2.85	27.3%	74.00
4	40,327	411.63	1.71	1.76	0.64	1.12	10.7%	27.27
5	44,372	421.85	1.28	0.93	0.47	0.47	4.5%	11.11

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,171	149.55	2.93	4.06	0.75	3.32	31.8%	221.76
2	32,362	347.80	2.17	3.98	1.04	2.93	28.1%	84.30

3	37,415	391.66	2.82	3.69	0.78	2.91	27.9%	74.34
4	40,575	413.50	1.82	1.82	0.58	1.24	11.8%	29.92
5	43,876	423.44	1.27	0.47	0.42	0.05	0.5%	1.18

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	77,748	665.87	-0.91	1.83	1.22	0.60	5.8%	9.09
41-60d	33,200	382.82	2.00	2.13	0.81	1.32	12.6%	34.43
61-90d	48,100	512.60	3.48	4.25	1.11	3.13	30.0%	61.13
>90d	14,356	164.75	6.44	5.82	0.44	5.39	51.6%	327.00

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	50,364	426.09	4.45	5.15	0.85	4.30	41.1%	100.86
2002-07	58,164	591.79	4.61	5.74	1.19	4.55	43.6%	76.91
2008-13	64,876	708.16	1.94	3.13	1.54	1.59	15.3%	22.53

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	32,133	320.84	2.65	1.97	0.66	1.31	12.5%	40.71
Chems	3,926	37.70	0.07	0.17	0.07	0.10	1.0%	27.76
Durbl	4,066	40.94	0.35	0.53	0.09	0.44	4.2%	107.52
Enrgy	9,095	85.98	0.87	1.08	0.16	0.91	8.7%	106.13
Hlth	17,555	165.49	1.56	1.52	0.38	1.14	10.9%	69.01
Manuf	17,051	173.75	0.65	1.30	0.35	0.94	9.0%	54.30
Money	25,815	263.22	1.47	2.73	0.54	2.19	21.0%	83.22
NoDur	7,510	73.82	0.38	0.44	0.14	0.30	2.9%	40.66
Other	26,581	265.30	1.49	2.17	0.60	1.57	15.0%	59.13
Shops	17,607	175.59	1.48	1.73	0.36	1.37	13.1%	77.81
Telcm	5,624	56.27	-0.05	0.18	0.11	0.07	0.7%	13.38
Utils	6,441	67.13	0.08	0.21	0.12	0.09	0.9%	14.04

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	57,238	573.05	1.82	2.58	1.06	1.52	14.5%	26.47
low vol	58,337	579.75	4.55	5.97	1.32	4.65	44.5%	80.16
mid vol	55,592	550.31	4.42	5.25	1.14	4.11	39.4%	74.71

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	3.88%	0.00%
0.5%	0.9	3.65%	0.23%
1.0%	1.8	3.41%	0.47%
2.0%	3.5	2.90%	0.98%
5.0%	8.8	1.01%	2.87%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.